| | |
|---|---|
| **Team Number:** | 104 |
| **Project Name:** | **HaMaLoYa** |
| **Team Member Names:** | **Hannah Ross, Matthew Hopwood, Logan Maelstron** |
| **Version:** | **Yazmin Herrera** |

***A. List ALL major components (active devices, integrated circuits, etc.) e***

| All Major Components | Component Name | Part Number |
|---|---|---|
| | PIC Microntroller | PIC18F57Q43 |
| | Speaker | N/A |
| | NPN Transistor | 2N3904 |
| | PNP Transistor | 2N3906 |
| | LED (red) | N/A |

***B. Assign each major component above to ONE power rail below. Try to***

| *+12V Power Rail* | Component Name | Part Number |
|---|---|---|

***c1. Regulator or Source Choice***

| *+5V Power Rail* | Component Name | Part Number |
|---|---|---|
| | PIC Microntroller | PIC18F57Q43 |
| | Speaker | N/A |
| | NPN Transistor | 2N3904 |
| | PNP Transistor | 2N3906 |
| | LED (red) | N/A |

***c2. Regulator or Source Ch*** +5V Regulator LM7805

| *-5V Power Rail* | Component Name | Part Number |
|---|---|---|

*c3. Regulator or Source Choice*

| *+3.3V Power Rail* | Component Name | Part Number |
|---|---|---|

*c4. Regulator or Source Choice*

***C. For each power rail above, select a specific voltage regulator using th***

***D. Select a specific external power source (wall supply or battery) for you***

| *External Power Source 1* | Component Name | Part Number |
|---|---|---|
| **Power Source 1 Selection** | Plug-in Wall Supply | Model:0930 |
| **Power Rails Connected to External Power Source 1** | +5V Regulator | LM7805 |

| *External Power Source 2* | Component Name | Part Number |
|---|---|---|
| **Power Source 2 Selection** | | |
| **Power Rails Connected to External Power Source 2** | | |

***E. Calculate Battery Life (if applicable). For each battery, also check the***

| | Component Name | Part Number |
|---|---|---|
| | Battery | |

***Notes***

External Supply Voltage should be determined by the dropout voltage for highe
If you have multiple units in your design (e.g., a base unit and remote unit) ther

# Power Budget

n

| xcept for power sources, voltage regulators, resistors, capacitors, or pas | | |
|---|---|---|
| **SupplyVoltageRange** | **#** | **AbsoluteMaximumCurrent (mA)** |
| +1.8V - 5.5V | 1 | 8.2 |
| +2.8V - 2.8V | 1 | 353 |
| +40V - 0V | 1 | 200 |
| +40V - 0V | 1 | 200 |
| +1.4V - 0V | 1 | 10 |

| minimize the number of different power rails in the design. | | |
|---|---|---|
| **SupplyVoltageRange** | **#** | **AbsoluteMaximumCurrent (mA)** |

***Subtotal***
***Safety Margin***
***Total Current Required on +12V Rail***

***Total Remaining Current Available on +12V Rail***

| SupplyVoltageRange | # | AbsoluteMaximumCurrent (mA) |
|---|---|---|
| +1.8V - 5.5V | 1 | 8.2 |
| +2.8V - 2.8V | 1 | 353 |
| +40V - 0V | 1 | 200 |
| +40V - 0V | 1 | 200 |
| +1.4V - 0V | 1 | 10 |

***Subtotal***
***Safety Margin***
***Total Current Required on +5V Rail***

| | | |
|---|---|---|
| +40V - 0V | 1 | 1000 |

***Total Remaining Current Available on +5V Rail***

| SupplyVoltageRange | # | AbsoluteMaximumCurrent (mA) |
|---|---|---|

*Subtotal*
*Safety Margin*
***Total Current Required on -5V Rail***

***Total Remaining Current Available on -5V Rail***

| SupplyVoltageRange | # | AbsoluteMaximumCurrent (mA) |
|---|---|---|

*Subtotal*
*Safety Margin*
***Total Current Required on +3.3V Rail***

***Total Remaining Current Available on 3.3V Rail***

***e same process as for major component selection. Confirm that the Total***

***ur system, and confirm that it can supply all of the regulators for all of the***

| SupplyVoltageRange | Output | AbsoluteMaximumCurrent (mA) |
|---|---|---|
| 100-240VAC | 9V | 3000 |
| +40V - 0V | 1 | 1000 |

***Total Remaining Current Available on External Power Source 1***

| SupplyVoltageRange | Output | AbsoluteMaximumCurrent (mA) |
|---|---|---|

***Total Remaining Current Available on External Power Source 2***

***worst-case lifetime of the battery by indicating the capacity in mAh.***

| SupplyVoltageRange | Capacity(mAh) |
|---|---|
| | 0 |

***Battery Life***

est-voltage regulator (e.g., +14V for a +12V regulator).
n you need a separate power budget for each unit

| ssive elements | |
|---|---|
| **TotalCurrent(mA)** | **Unit** |
| 8.2 | mA |
| 353 | mA |
| 200 | mA |
| 200 | mA |
| 10 | mA |
| | mA |

| | |
|---|---|
| **TotalCurrent(mA)** | **Unit** |
| 0 | mA |
| 0 | mA |
| 0 | mA |
| 0 | mA |
| 0 | mA |
| 0 | mA |
| 25% | |
| 0 | mA |
| | |
| 0 | mA |
| 0 | mA |
| **TotalCurrent(mA)** | **Unit** |
| 8.2 | mA |
| 353 | mA |
| 200 | mA |
| 200 | mA |
| 10 | mA |
| 771.2 | mA |
| 25% | |
| 964 | mA |
| | |
| 1000 | mA |
| 36 | mA |
| **TotalCurrent(mA)** | **Unit** |
| 0 | mA |
| 0 | mA |
| 0 | mA |
| 0 | mA |

| | |
|---|---|
| 0 | mA |
| 25% | |
| 0 | mA |
| | |
| | mA |
| 0 | mA |

| TotalCurrent(mA) | Unit |
|---|---|
| 0 | mA |
| 0 | mA |
| 0 | mA |
| 0 | mA |
| 0 | mA |
| 25% | |
| 0 | mA |
| | |
| | mA |
| 0 | mA |

***Remaining Current Available***

***e power rails simultaneously. If***

| TotalCurrent(mA) | Unit |
|---|---|
| 3000 | mA |
| | |
| 0 | mA |
| 1000 | mA |
| 0 | mA |
| 2000 | mA |

| TotalCurrent(mA) | Unit |
|---|---|
| 0 | mA |
| | |
| 0 | mA |
| | |
| | |
| 0 | mA |

**equiredByRegulators**

| | |
|---|---|
| 0 | |
| #DIV/0! | hours |